Yash Pratap Singh

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SKILLS

- **Languages:** Java, Python, C, C++, SQL
- **Libraries/Frameworks:** NumPy, Pandas, Scikit-learn, Matplotlib, HTML, CSS, Javascript
- **Tools/Platforms:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Cisco Packet Tracer, Linux
- **Soft Skills:** Problem-Solving, Analytical Thinking, Team Collaboration, Continuous Learning, Cross-Functional Communication, Adaptability

INTERNSHIP

Machine Learning Intern | YHills Edutech | Certificate Apr 2025

- Developed and evaluated supervised machine learning models in Python using NumPy, Pandas, and Scikit-Learn, achieving robust baseline accuracy across tabular datasets.
- Implemented end-to-end data preprocessing pipelines, covering automated missing value imputation, outlier detection, and categorical feature encoding to improve data quality.
- Optimized model evaluation workflows using cross-validation and hyper parameter tuning, standardizing performance benchmarking against precision, recall, and ROC-AUC metrics.

PROJECTS

Role-Based Hospital Management System (MediFlow) | GitHub. Jul 2026

- Architected an enterprise-grade hospital management portal in Java implementing strict Role-Based Access Control (RBAC) across 7 distinct actors (Admin, Doctor, Patient, Pharmacist, Cashier, Lab Specialist, Receptionist) to enforce end-to-end medical data privacy.
- Engineered a decoupled, multi-department pipeline managing state transitions across patient registration, clinical diagnosis, dynamic lab test routing, and automated multi-tier billing calculations.
- Applied core Object-Oriented Design (OOP) principles—including polymorphism, interfaces, and composition alongside Java Collections and custom exception handling to ensure data integrity and zero redundant record duplication across workflows.

Tech: Java, Object-Oriented Programming (OOP), Data Structures & Collections, RBAC, HTML/CSS, JavaScript, Git

MedPredict: Clinical Health Risk Prediction Platform | GitHub Apr 2025

- Architected the backend ML data pipeline across 4 clinical disease modules (Obesity, Heart Disease, Sleep Apnea, Diabetes), implementing data deduplication (723 rows removed), imputation, and target leakage prevention.
- Handled clinical class imbalance using SMOTE and trained ensemble architectures (XGBoost, Random Forest), achieving **84.93% accuracy** (84.39% F1-score) on Obesity classification and **78.69% accuracy** on Heart Disease screening.
- Evaluated cross-validation performance via 10-fold Stratified Cross-Validation and engineered an optimized inference pipeline with custom probability thresholding (0.3 threshold) to maximize clinical recall for high-risk detection.

Tech: Python, Scikit-Learn, XGBoost, Random Forest, SMOTE, Pandas, NumPy, 10-Fold Cross-Validation, Flask

TRAINING

Data Structures and Algorithms using Java Programming | CipherSchools | Certificate Aug 2026

- Completed intensive core computer science training, mastering Data Structures, Algorithms, and Object Oriented Programming (OOP) paradigms in Java.
- Solved algorithmic challenges focused on time-space complexity optimization (O(n log n)) across trees, graphs, dynamic programming, and recursion.
- Engineered a role-based hospital management capstone using Java Collections, abstract interfaces, and robust exception handling to simulate enterprise workflows.

CERTIFICATES

- Oracle Cloud Infrastructure Certified AI Foundations Associate | Oracle | Certificate — Sep 2026
- Database Management System Part – 1 | Infosys Springboard | Certificate — Jul 2026
- Programming in JAVA | Neo Colab (LPU & iamneo) | Certificate — May 2026
- Machine Learning with Python | YHills | Certificate — Apr 2025
- Linux Commands and Shell Scripting | Skillera | Certificate — Nov 2024

ACHIEVEMENTS

- *Serving as Head of Technical Team — Accenture Emph Club | LPU CPE* | Link — Jul 2025 - Present
- *EarnedSecured "Top Performer" & Contest Winner among 500+ participants — Tech Synergy Coding Contest | CipherSchools* | Certificate — Jul 2026
- *Secured 3rd Position among 300+ CS students — Byte Battle 1.0: Technical Quiz Competition | Lovely Professional University* | Certificate — Nov 2024
- *Secured Top 30 Rank among 1,000+ engineering participants — CodeXtreme 2.0 Neo Codeathon | iamneo & LPU* | Certificate — Apr 2024

CO-CURRICULAR ACTIVITIES

- Community Outreach Intern — CyberSmart Awareness (WNS Cares Foundation) | Certificate — Sep 2025
- Computer Programming Trainee (72-Hour MOOC) — iamneo Platform & LPU | Certificate — May 2025
- Advanced Data Analysis Participant — Data Made Easy: Excel Workshop | LPU CPE | Certificate — Feb 2025

EDUCATION

Lovely Professional University Phagwara, Punjab
Bachelor of Technology — Computer Science and Engineering | CGPA: 8.62 Aug 2024 – Present

Maria School Jaspur, Uttarakhand
Intermediate; Percentage: 73.5% Apr 2022 – Mar 2023

Maria School Jaspur, Uttarakhand
Matriculation; Percentage: 87.6% Apr 2020 – Mar 2021